

MUSTHOFA JOKO ANGGORO

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RESEARCH INTERESTS

Computer systems architecture, distributed systems, operating systems, hardware-software co-design — with particular interest in edge AI systems, resource-constrained computing, and energy-efficient system design across the computing continuum

EDUCATION

University of Indonesia (UI) Jakarta, Indonesia
Bachelor of Computer Science, Information Systems *Expected Graduation: Aug 2026*

- **GPA: 3.84/4.00**
- **Thesis:** Implementation of Morris Mano Computer Architecture with Enhancements on Xilinx Spartan-3 FPGA (X3700AN) using VHDL — demonstrating hardware design, digital logic optimization, and system-level architecture implementation
- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

Nanyang Technological University (NTU) & ITS Singapore & Surabaya
INSPIRASI-NTU Summer Program 2025 (LPDP Scholarship Awardee) *July 2025*

- Conducted sustainability-focused research on smart port automation and green technology adoption in distributed maritime systems, analyzing IoT sensor networks, automation architectures, and energy-efficient deployments
- Produced comprehensive technical report proposing tiered port automation framework emphasizing renewable energy utilization and carbon emission reduction — directly aligned with sustainability objectives in edge computing systems
- Collaborated with international research teams on applying systems thinking to environmental challenges

EXPERIENCE

Software Engineer Intern Traveloka (Travel Technology Company)
Systems Engineering & Backend Development *Aug 2025 - Present*

- **Edge AI-Relevant Skills:** Resource optimization, distributed system architecture, fault tolerance, cloud infrastructure (AWS, Terraform), containerization (Docker, Kubernetes), performance tuning, reliability engineering

SKILLS

Systems Programming: C, Python, Java, VHDL (hardware description), Go

Hardware & Architecture: FPGA development (Xilinx Spartan-3), VHDL, digital logic design, computer architecture implementation, hardware-software co-design

Machine Learning: Scikit-learn, TensorFlow, supervised/unsupervised learning, model evaluation, imbalanced learning

Distributed Systems: Microservices architecture, containerization (Docker, Kubernetes), cloud infrastructure (AWS, Terraform)

Operating Systems: Linux, Unix — OS internals, process management, memory management, system-level programming

MACHINE LEARNING PROJECTS

Used Car Sales Prediction Course Project - AI & Data Science
Machine Learning Analyst *Apr 2025 - May 2025*

- Built comprehensive predictive modeling system for used car sales using imbalanced classification, regression, and clustering techniques to derive actionable business insights from complex datasets
- Implemented multiple ML algorithms (Gradient Boosting, Random Forest, Logistic Regression, Linear Regression, Naive Bayes, K-Means) with emphasis on handling imbalanced data through oversampling and undersampling techniques
- Conducted extensive exploratory data analysis (EDA) and feature engineering, achieving robust model performance through systematic preprocessing, evaluation, and optimization using Scikit-learn
- **Tools & Techniques:** Python, Scikit-learn, Pandas, NumPy, imbalanced learning methods, model evaluation metrics